Prediction Markets Are Fisher Markets

Batch Auction Clearing via Eisenberg-Gale Duality

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Summary. We identify the exact reduced-form utility of a budgeted market maker with retained cash (Proposition 3): affine when capital is slack and logarithmic when capital binds. This converts prediction market batch auctions into a Fisher-market clearing program (Theorem 5). A deployed-value lift $V_k = U_k + s_k$ derives the reduced form exactly and yields a concave batch-clearing program with exact KKT interpretation and polynomial-time conic formulations. Budget constraints vanish from the formulation entirely, absorbed into the objective. For every $b > 0$, clearing prices are unique; at $b = 0$, prices can be non-unique as in the LP. Prices, fills, and capital deployment emerge simultaneously from a single optimization, although order-level fills need not be unique without additional nondegeneracy assumptions.

1. Introduction

The main result of this paper is that prediction market batch auctions admit a Fisher-market formulation. The decisive observation is that a budgeted MM with retained cash has a simple reduced-form utility: affine when capital is slack, logarithmic when capital binds.

A prediction market exchange runs *batch auctions*: orders accumulate, then clear simultaneously at uniform prices. The clearing problem is an allocation problem — find the fills and prices that maximize welfare subject to balance constraints. Without budget constraints, this is a standard Linear Program.

Market makers introduce the main difficulty. An ordinary participant submits a single order: "buy 100 shares of Yes at \$0.60." The capital required (\$60) is known at submission and locked upfront — no budget constraint needed. A market maker, by contrast, posts *hundreds* of orders across *dozens* of markets simultaneously. The total capital consumed depends on which orders fill and at what prices — both determined by the auction. The MM deposits a finite balance B_k and the exchange must ensure total spending does not exceed it. This budget constraint is *bilinear*: capital consumed depends on the clearing price, and the clearing price depends on which fills happen. This single constraint makes the feasible set non-convex.

The key object is therefore not the hard budget constraint but the MM's reduced-form utility. Let U_k be the weighted fill obtained by MM k. Once retained cash is allowed, optimizing over that cash induces a utility Proposition 3 that is affine below budget and logarithmic above it. The deployed-value variable $V_k = U_k + s_k$ is the technical lift that derives this reduced form exactly and later makes the KKT system and conic formulation transparent. This single observation explains the whole paper: LP recovery when budgets do not bind, endogenous budget absorption when they do, and a clean price dual.

We prove three things, in order:

1. **The framework (§2).** LP batch clearing and Hanson's LMSR are the same mathematical object at different temperatures, connected by Fenchel duality. This is the foundation for everything that follows. It is also where we prove that clearing prices are unique when budgets are absent — a fact that makes the budget obstacle precise.

2. **The obstacle (§3).** Adding budget constraints to the risk-neutral model can make the feasible set non-convex. Standard convex optimization no longer applies, no polynomial-time algorithm is known, and uniqueness of clearing prices is an open question. The structural cause: linear welfare with hard budget caps is an internally inconsistent economic model.
3. **The resolution (§§4–6).** Retained cash induces an exact MM utility Proposition 3: affine when capital is slack, logarithmic when capital binds. This reduced form yields a concave batch-clearing program, exact limit-order KKT conditions, endogenous budget absorption, unique smoothed prices for $b > 0$, and polynomial-time conic formulations. The deployed-value lift is the proof and solver layer behind this reduced form. This is the Fisher market isomorphism.

The modeling "trick" is not merely logarithmic utility — it is identifying the reduced-form MM utility induced by budget plus retained cash. When we replace the linear-with-hard-cap model by that reduced form, the pathology fixes itself. §6 shows the result is practical: the program admits a conic formulation with K exponential cones and suggests a smooth budget-allocation decomposition across independent market groups.

2. Foundations: Batch Clearing and LMSR

This section builds the mathematical framework for the main result. Readers familiar with cost-function market makers (Abernethy et al. 2013) will recognize these results in a new framing.

2.1. Minting Cost and the LP

Consider a prediction market with K mutually exclusive outcomes. In a batch auction, N orders arrive. Each order i has a limit price L_i, a maximum quantity $\overline{Q}_i$, a side (buy or sell), and a target market $m(i)$. Some orders belong to market makers; we write MM_k for the set of orders belonging to MM k.

The *net demand* — excess buy over sell volume — for each outcome is:

$$D_k = \sum_{i \in \text{ buy}(k)} q_i - \sum_{j \in \text{ sell}(k)} q_j$$

where $q_i \in \left[0, \overline{Q}_i\right]$ is the fill quantity of order i.

To clear the market, the exchange *mints* complete sets: one share of every outcome at cost \$1 (exactly one resolves to \$1 at settlement, so this is fairly priced). To supply D_k net shares of each outcome, exactly $\max_k D_k$ mints are needed — this covers the highest-demand outcome, with surplus shares of other outcomes left over. The minting cost is therefore $V(\boldsymbol{D}) = \max_k D_k$, and the welfare-maximizing clearing solves:

$$P: \quad \max_{\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]} \quad \sum_i w_i q_i - \max_k D_k(\boldsymbol{q})$$

where w_i is the welfare coefficient of order i ($+L_i$ for buyers, $-L_i$ for sellers) and $\overline{\boldsymbol{Q}} = \left(\overline{Q}_1, ..., \overline{Q}_N\right)$ is the vector of max fill quantities. Since V is convex and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$, the objective is concave — this is a convex optimization problem. Introducing an explicit minting variable $M \geq D_k$ for each k, this is equivalent to the Linear Program:

$$\max_{\boldsymbol{q}, M} \quad \sum_i w_i q_i - M \quad \text{s.t.} \quad D_k(\boldsymbol{q}) \leq M \quad \forall k, \quad \boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right]$$

Its dual variables are clearing prices.

2.2. Fenchel Duality: Prices from Conjugates

The constraint that clearing prices must be probabilities ($\sum p_k = 1,\ p_k \geq 0$) is not imposed by fiat — it is the Fenchel conjugate of the minting cost. Minting at \$1 per complete set *encodes* the probability axiom.

Theorem 1 (Minting–Simplex Duality). *The Fenchel conjugate of $V(\boldsymbol{D}) = \max_k D_k$ is the indicator function of the probability simplex:*

$$V^*(\boldsymbol{p}) = \delta_\Delta(\boldsymbol{p}) = \begin{cases} 0 & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

where $\Delta = \left\{\boldsymbol{p} \geq 0 : \sum_k p_k = 1\right\}$.

Proof. For $\boldsymbol{p} \in \Delta$: convexity gives $\sum p_k D_k \leq \max_k D_k$, so the supremum is 0 (attained at $\boldsymbol{D} = \boldsymbol{0}$). For $\boldsymbol{p} \notin \Delta$: some deviation from the simplex exists, and scaling $\boldsymbol{D}$ in the exploiting direction sends the supremum to $+\infty$. □

The probability axiom $\sum p_k = 1$ is not a modeling choice — it is a no-arbitrage condition. Any deviation from the simplex is an arbitrage opportunity (e.g., if $\sum p_k > 1$, mint at \$1 and sell at $\sum p_k$), and the conjugate enforces this as a hard constraint.

2.3. Entropy Smoothing: LMSR as Soft LP

Hanson's LMSR cost function $C_{b(\boldsymbol{D})} = b \ln \sum \exp(D_k/b)$ is a smooth (C^∞) version of the minting cost, parametrized by a temperature $b > 0$. The structural content is in its Fenchel conjugate: where $V^* = \delta_\Delta$ was a hard constraint (prices must be probabilities), C_b^* is a soft penalty (prices near uniform are cheap, prices near a vertex are expensive).

Theorem 2 (LMSR–Entropy Duality). *The Fenchel conjugate of C_b is negative Shannon entropy on the simplex:*

$$C_b^*(\boldsymbol{p}) = \begin{cases} b \sum_k p_k \ln p_k & \text{if } \boldsymbol{p} \in \Delta \\ +\infty & \text{otherwise} \end{cases}$$

Proof. We compute $C_b^*(\boldsymbol{p}) = \sup_{\boldsymbol{D}}\{\sum p_k D_k - b \ln \sum \exp(D_k/b)\}$ by setting the gradient to zero:

$$p_k - \frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)} = 0 \quad \Rightarrow \quad p_k = \frac{\exp(D_k/b)}{\sum_j \exp(D_j/b)}$$

This is the *softmax* — exactly the LMSR marginal price. Inverting: $D_k = b \ln p_k + b \ln Z$ where $Z = \sum \exp(D_j/b)$. Substituting back:

$$\langle \boldsymbol{p}, \boldsymbol{D} \rangle = b \sum_k p_k \ln p_k + b \ln Z, \qquad C_b(\boldsymbol{D}) = b \ln Z$$

So $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k$. This is finite only when $\boldsymbol{p} \in \Delta$ (the softmax always yields a probability vector). □

The approximation quality is controlled by b:

Proposition 1 (LSE–Max Sandwich). $\max_k D_k \leq C_{b(\boldsymbol{D})} \leq \max_k D_k + b \ln K$. *The gap $b \ln K$ is the maximum LMSR subsidy.*

Proof. $\exp(\max_k D_k/b) \leq \sum \exp(D_k/b) \leq K \exp(\max_k D_k/b)$. Apply $b \ln$. □

Choosing b in practice. The parameter b is a smoothing knob, and Proposition 1 gives its dollar-scale cost directly: the maximum extra minting subsidy is $b \ln K$. A practical rule is to choose b so that $b \ln K$ is below the venue's tolerated per-batch subsidy or below one economically meaningful price tick. Then the smoothed auction remains close to LP clearing while still delivering unique clearing prices for every $b > 0$.

The complete picture:

$$\begin{array}{ccc} V = \max_k D_k & \xleftrightarrow{\text{Fenchel}} & V^* = \delta_\Delta \\ \uparrow\ b \to 0 & & \uparrow\ b \to 0 \\ C_b = b \ln \sum \exp(D_k/b) & \xleftrightarrow{\text{Fenchel}} & C_b^* = b \sum p_k \ln p_k \end{array}$$

As $b \to 0$, the smooth cost C_b sharpens to the LP cost V, and the entropy penalty hardens to the simplex indicator.

2.4. The Smoothed Batch Auction and Its KKT Conditions

Replace the minting cost V with C_b in the batch clearing:

$$P_b : \quad \max_{\boldsymbol{q} \in [0, \overline{\boldsymbol{Q}}]} \quad \sum_i w_i q_i - C_{b(\boldsymbol{D}(\boldsymbol{q}))}$$

Since C_b is convex and smooth, and $\boldsymbol{D}(\boldsymbol{q})$ is linear, P_b is a smooth concave maximization. Its first-order conditions are necessary and sufficient, and the exponentials in the clearing prices come from the minting cost, not the order book.

Theorem 3 (LMSR = Smoothed Batch Clearing). *At the optimum of P_b, the clearing prices are the softmax of net demand:*

$$p_k^* = \frac{\partial C_b}{\partial D_k} = \frac{\exp(D_k^*/b)}{\sum_j \exp\left(D_j^*/b\right)}$$

This is the LMSR marginal price function. By construction, $\sum_k p_k^* = 1$.

The two limits are immediate: as $b \to 0$, $P_b \to P$ (by Berge's theorem on the compact feasible set, with uniform convergence from Proposition 1); as $b \to \infty$, $\exp(D_k/b) \to 1$ for all k, so $p_k \to 1/K$.

The KKT conditions yield the standard *Uniform Clearing Price* (UCP) rule: order i fills if and only if its limit price exceeds the clearing price p_k. The entropy smoothing does not alter order-matching logic; it only changes how prices depend on quantities: continuously via softmax rather than discretely via the marginal order.

2.5. Price Uniqueness Without Budgets

Without budget constraints, clearing prices are *always* unique — for any $b > 0$, any order book, unconditionally. This is proved by passing to the Fenchel dual, where the entropy term provides strict convexity. Write $W(\boldsymbol{D})$ for the maximum welfare achievable at a given demand vector $\boldsymbol{D}$ (the inner LP over fills $\boldsymbol{q}$ with $\boldsymbol{D}$ fixed). The primal problem in demand space is $\max_{\boldsymbol{D}}[W(\boldsymbol{D}) - C_b(\boldsymbol{D})]$, where W is concave piecewise-linear. By Fenchel-Rockafellar duality, this is equivalent to minimizing over prices:

$$\min_{\boldsymbol{p} \in \Delta} \left[\underbrace{W^*(\boldsymbol{p})}_{\text{consumer surplus}} + \underbrace{C_b^*(\boldsymbol{p})}_{\text{entropy penalty}} \right]$$

where $W^*(\boldsymbol{p})$ is the total surplus of orders whose limit prices exceed clearing prices, and $C_b^*(\boldsymbol{p}) = b \sum p_k \ln p_k$ is the negative Shannon entropy (Theorem 2).

Theorem 4 (Unconstrained Price Uniqueness). *The Fenchel dual of the unconstrained smoothed problem is strictly convex on the simplex: W^* is convex and $C_b^*(\boldsymbol{p}) = b\sum_k p_k \ln p_k$ is strictly convex on Δ (with the convention $0 \ln 0 := 0$). The clearing prices $\boldsymbol{p}^*$ are therefore unique for any $b > 0$ and any order book, unconditionally.*

Proof. W^* is convex because it is a Fenchel conjugate. The scalar function $x \mapsto x \ln x$ is strictly convex on $[0, +\infty)$, so $\sum_k p_k \ln p_k$ is strictly convex on the convex set Δ. Therefore $W^* + C_b^*$ is strictly convex on Δ. Since Δ is compact and the dual objective is lower-semicontinuous, a minimizer exists; strict convexity implies it is unique. □

3. The Budget Obstacle

When market makers have budget constraints, the clearing problem becomes fundamentally harder. Without budgets, batch clearing is a standard convex program (Theorem 4). With budgets, the capital constraint is bilinear — capital consumed depends on clearing prices, which depend on fills — and standard convex optimization no longer applies.

3.1. The Bilinear Constraint

A market maker k deposits balance B_k and posts orders across multiple markets. The capital consumed by each fill depends on the clearing price:

$$\mathrm{cap}_{k(\boldsymbol{p},\boldsymbol{q})} = \sum_{i \in \mathrm{MM}_k} c_{i\left(p_{m(i)}\right)} \cdot q_i, \quad c_{i(p)} = \begin{cases} p & \text{if BuyYes/SellNo} \\ 1 - p & \text{if SellYes/BuyNo} \end{cases}$$

The budget constraint $\mathrm{cap}_k \leq B_k$ is *bilinear*: p is determined by $\boldsymbol{q}$ through the clearing mechanism. The product $c(p(\boldsymbol{q})) \cdot q$ can make the feasible set non-convex in fill space. This single constraint is what separates prediction market clearing from a standard LP.

3.2. Computational Consequences

The bilinear budget constraint has three consequences for computation:

1. The feasible set can be non-convex. Define $h_k(\boldsymbol{q}) = \sum_{i \in \mathrm{MM}_k} p_{m(i)}(\boldsymbol{q}) \cdot q_i$ as MM k's capital consumption. We start with the simplest non-trivial case: an MM with buy orders on two independent binary groups, one order per group, fill quantities q_1, q_2. Since groups are independent, $h = f(q_1) + f(q_2)$ where $f(q) = \sigma(q/b) \cdot q$ and σ is the logistic sigmoid. Differentiating:

$$f''(q) = \frac{\sigma(1-\sigma)}{b}\left[2 + \frac{q(1-2\sigma)}{b}\right]$$

Since $\sigma(1-\sigma) > 0$, the sign depends on $2 + q(1-2\sigma)/b$. At $q = 0$: $f''(0) = 1/(2b) > 0$ (convex). For large q: $\sigma \to 1$ and the bracket becomes $2 - q/b \to -\infty$ (concave). The inflection point $f''(q^*) = 0$ is at $q^* \approx 2.4b$. So h is neither convex nor concave.

This witness is not knife-edge in b. Writing $f_{b(q)} = \sigma(q/b)q$, we have the scaling identity $f_{b(bt)} = bf_1(t)$ and therefore $h_{b(\boldsymbol{q})} = bh_1(\boldsymbol{q}/b)$. So for *every* $b > 0$, the points

$$x_b = (2b, 9b), \quad y_b = (9b, 2b), \quad m_b = (5.5b, 5.5b)$$

satisfy

$$h_{b(x_b)} = h_{b(y_b)} \approx 10.7605b, \quad h_{b(m_b)} \approx 10.9552b$$

Hence for budget $B = 10.8b$, both endpoints satisfy $h_b \leq B$ while the midpoint violates it. Therefore the sublevel set $\{h_b \leq B\}$ is non-convex for every $b > 0$.

2. No standard convex algorithm applies directly. The unconstrained problem P_b is a smooth concave maximization (solved by interior point in polynomial time). Adding budget constraints $\{h_k \le B_k\}$ can make the feasible set non-convex. Standard convex optimization — interior point, projected gradient, Frank-Wolfe — requires convex feasible sets. The budget-constrained problem is a bilinear program, for which no polynomial-time algorithm is known in general.

3. The uniqueness question is open. The proof of Theorem 4 (price uniqueness without budgets) uses strict convexity of the Fenchel dual. With budgets, the argument breaks: uniqueness at one price vector requires *cross-price budget feasibility* ($\mathrm{cap}_k(\boldsymbol{p}^2, \boldsymbol{q}^1) \le B_k$), which the bilinear constraint does not guarantee. Whether clearing prices are nonetheless unique for all LMSR instances is an open question. (In the degenerate case of two identical markets with a symmetric MM, two KKT points exist by symmetry — but this requires exact parameter tuning and does not extend to generic order books.)

> **Proposition 2** (Computational Obstruction). *For every $b > 0$, the budget-constrained risk-neutral clearing problem admits instances whose feasible set $\{\boldsymbol{q} \in \mathcal{C} : h_k(\boldsymbol{q}) \le B_k\}$ has non-convex sublevel sets, so the problem is a bilinear program and standard convex optimization does not apply directly. By contrast, the risk-averse program P_b^{RA} (Theorem 5) is a standard convex program with conic formulations solvable in polynomial time.*

In practice, exchanges handle budgets by iterative heuristics: solve the LP ignoring budgets, check for violations, adjust, repeat. Such methods have no convergence guarantee and can cycle. The risk-averse program eliminates this entirely — budgets are absorbed into the objective.

3.3. The Expenditure Perspective

A natural idea: change variables to capital expenditures $e_i = c_{i(p)} \cdot q_i$, which linearizes the budget constraint to $\sum_{i \in \mathrm{MM}_k} e_i \le B_k$. In Fisher markets, the Eisenberg-Gale program works precisely because this change of variables produces a convex program. Does it work here?

No. Welfare transforms to $\sum_i w_i \cdot e_i / c_{i(p)}$ — a *rational* function of prices. In Fisher markets (Eisenberg & Gale 1959), the analogous program is convex because agents have *diminishing-returns* utilities: $\sum B_k \ln U_k$ provides curvature. Our MMs have constant marginal returns (linear welfare), providing none. The expenditure substitution linearizes the budget but introduces non-convexity into the objective.

The diagnosis is structural: *linear welfare with hard budget caps* is an internally inconsistent economic model — agents simultaneously risk-neutral (constant marginal value) and risk-averse (capped exposure). The non-convexity is the computational symptom.

The mathematics tells us exactly what is needed: a concave reduced-form MM objective with endogenous cash retention. §4 argues that diminishing returns is not merely a mathematical convenience but a plausible model for repeat-participation MMs. §5 derives the resulting utility, and §6 uses it to build the clearing program.

4. Why Diminishing Returns Is a Plausible MM Model

The previous section showed that clearing requires diminishing-returns utility to be computationally tractable. This section is a modeling argument rather than part of the proof spine: it gives reasons diminishing returns is a plausible reduced-form model for repeat-participation MMs.

4.1. Kelly as Repeated-Game Motivation

Breiman (1961) showed that Kelly-style log utility is growth-optimal under classical repeated favorable-bet assumptions. This does not imply that every real MM literally maximizes log utility, but it does provide a principled repeated-game foundation for diminishing returns.

A prediction market exchange runs *repeated* batch auctions, compounding returns. If an MM evaluates batches through long-run capital growth, an objective of the form $\max \sum_t \mathbb{E}[\ln(1 + r_t)]$ is natural. Linear utility remains appropriate for one-shot welfare, but it misses sizing and survival tradeoffs across batches.

The modeling claim is therefore modest: log utility is not arbitrary; it is a defensible reduced form for repeat participants who care about growth and drawdown.

4.2. Budget Limits Encode Risk Aversion

Why do MMs have budget constraints at all? A risk-neutral agent with positive expected value should bet everything. The existence of $B_k < \text{total wealth}$ is itself evidence of risk aversion: the MM limits exposure because it values capital preservation.

Linear welfare with a hard budget is internally inconsistent: "I value each share equally (risk-neutral) but I won't risk more than B_k (risk-averse)." The budget is a crude piecewise-linear approximation of what log utility handles smoothly.

4.3. Empirical Grounding

In practice, many MMs post *ladders* — multiple orders at decreasing limit prices with decreasing quantities — revealing concave demand directly through the order book. They also impose position limits to cap inventory risk: a linear-utility MM is indifferent between the 1st and 1,001st share, but real MMs are not. Both ladders and position limits are piecewise-linear approximations of diminishing returns. Log utility captures concave demand natively and penalizes concentration intrinsically.

4.4. Synthesis

These arguments are motivational, not part of the theorem spine. The paper does *not* need Kelly to be the only correct behavioral model of real MMs. It only needs a concave MM objective with endogenous cash retention, and log utility gives the cleanest reduced form. The question is therefore: what utility does a budgeted MM with retained cash actually have? The answer is the affine-to-log function derived in the next section. The non-convexity of §3 is better viewed as a pathology of the linear-with-hard-caps model than as an inherent feature of prediction markets.

5. Reduced-Form Utility and Fisher-Market Clearing

Replacing linear MM welfare with Kelly-style diminishing returns transforms the budget-constrained clearing problem into a concave optimization problem. The main object is the reduced-form MM utility Proposition 3. The deployed-value variable V_k is the exact lift behind it: useful for derivation, KKT analysis, and conic computation, but not the main conceptual payload. Once the reduced form is identified, LP recovery, welfare-gap control, and the decomposition gradient all become consequences of one function.

5.1. Reduced-Form MM Utility

Proposition 3 (Reduced-Form MM Utility). *Optimizing over the deployed-value variable yields the MM utility*

$$\psi_{B_k}(U) = \begin{cases} U + B_k \ln B_k - B_k & \text{if } U \le B_k \\ B_k \ln U & \text{if } U \ge B_k \end{cases}$$

This utility is concave and C^1, with derivative $\psi_{(B_k)'}(U) = \min(1, B_k/U)$. It satisfies the affine-envelope bound $\psi_{B_k}(U) \le U + B_k \ln B_k - B_k$, with equality exactly on the slack-budget region $U \le B_k$. So MM utility is affine when budget is slack and logarithmic when capital binds.

Proof. Given fill value U, maximize over deployed value:

$$\psi_{B_k}(U) = \max_{V \ge U} [B_k \ln V - V + U]$$

The derivative and curvature are $B_k/V - 1$ and $-B_k/V^2 < 0$, so the maximizer is $V^* = \max(U, B_k)$. Substituting V^* gives the piecewise formula, and the derivative follows from the two branches. □

This is the paper's reduced form. LP recovery, the welfare-gap bound, and the decomposition gradient in the companion note are all consequences of the two regimes encoded by ψ_B: affine below budget, logarithmic above it.

Proposition 4 (Deployed-Value Lift). *The reduced-form clearing problem*

$$\max_{\boldsymbol{q} \in \mathcal{C}} \quad \sum_k \psi_{B_k}\big(U_{k(\boldsymbol{q})}\big) + \sum_{j \notin \text{ MM}} w_j q_j - C_{b(\boldsymbol{D}(\boldsymbol{q}))}$$

is equivalent to the deployed-value lift

$$\max_{\boldsymbol{q} \in \mathcal{C}, \boldsymbol{V} \ge \boldsymbol{U}(\boldsymbol{q})} \quad \sum_k \big[B_k \ln V_k - V_k + U_{k(\boldsymbol{q})}\big] + \sum_{j \notin \text{ MM}} w_j q_j - C_{b(\boldsymbol{D}(\boldsymbol{q}))}$$

because each MM term is the pointwise maximum over $V_k \ge U_k$ of the lifted objective.

Proof. Immediate from Proposition 3, applied MM-by-MM. □

5.2. Reduced-Form Clearing Program

Theorem 5 (Reduced-Form Clearing with Budgeted Market Makers). For each MM k, let ψ_{B_k} be the reduced-form utility of Proposition 3 and consider

$$P_b^{\mathrm{RA}}: \quad \max_{\boldsymbol{q}\in\mathcal{C}} \quad \underbrace{\sum_k \psi_{B_k}\left(U_{k(\boldsymbol{q})}\right)}_{\text{MM welfare}} + \underbrace{\sum_{j\notin \text{ MM}} w_j q_j}_{\text{retail welfare}} - \underbrace{C_{b(\boldsymbol{D}(\boldsymbol{q}))}}_{\text{minting cost}}$$

where MM_k^+ is the set of MM k's buy orders (sell orders from MMs enter the retail welfare term; see Buy/Sell Reduction below), $U_{k(\boldsymbol{q})} = \sum_{i\in \mathrm{MM}_k^+} L_i q_i \geq 0$ is MM k's total weighted fill ($L_i > 0$ for all $i \in \mathrm{MM}_k^+$), $\mathcal{C} = \left\{\boldsymbol{q} \in \left[0, \overline{\boldsymbol{Q}}\right] : \text{balance constraints}\right\}$ is the feasible fill polytope, and C_b is the smoothed minting cost (Proposition 1). At any optimum $\boldsymbol{q}^*$, define deployed value $V_k^* = \max\left(U_{k(\boldsymbol{q}^*)}, B_k\right)$ and capital-scarcity factor $\alpha_k = B_k / V_k^* = \psi_{(B_k)'}\left(U_{k(\boldsymbol{q}^*)}\right) \leq 1$. Then:

1. The objective is concave, the feasible set is convex, and an optimum exists.
2. The reduced-form program is equivalent to the deployed-value lift of Proposition 4.
3. Limit orders are exact: if MM order i fills, then $\alpha_k L_i \geq p_{m(i)}$, hence $L_i \geq p_{m(i)}$. No negative-welfare MM fill is possible.
4. No explicit budget constraints appear. Yet at the optimum, each MM k spends at most B_k: capital on fills plus retained cash $\sum_{i\in\mathrm{MM}_k^+} p_{m(i)} q_i + \left(V_k^* - U_{k(\boldsymbol{q}^*)}\right) \leq B_k$.
5. The program operates in two regimes per MM. If $U_{k(\boldsymbol{q}^*)} < B_k$, then $\alpha_k = 1$ and the MM clears exactly as in the risk-neutral LP. If $U_{k(\boldsymbol{q}^*)} > B_k$, then $\alpha_k = B_k / U_{k(\boldsymbol{q}^*)} < 1$ and lower-ROI fills are throttled.
6. For $b > 0$, clearing prices are unique and equal the softmax rule $p_k = \frac{\partial C_b}{\partial D_k}$. At $b = 0$, clearing prices are LP dual variables and may be non-unique.
7. The program is polynomial-time solvable through the conic reformulations in the Computation section.

Proof Map.

Move	**Object**	**Payoff**
Identify utility	Proposition 3	Affine-below-budget, log-above-budget MM objective
Lift to V	Proposition 4	Exact derivation, KKT system, and solver form
Read KKT	$\alpha_k = B_k / V_k$	Exact limit orders and budget absorption
Project to demand	Proposition 5 and $W^{\mathrm{RA}(\boldsymbol{D})}$	Concave demand-space objective
Dualize prices	Proposition 6	Unique clearing prices for every $b > 0$
Lift to cones	Exponential-cone reformulation	Polynomial-time solvability

The next auxiliary proposition is used only in part (4) of the proof, where the reduced-form clearing program is projected into demand space and dualized over prices.

Proposition 5 (Demand-Space Concavity). *Define the feasible demand polytope*

$$\mathcal{D} = \{\boldsymbol{D} : \exists \boldsymbol{q} \ \text{ with } \ \boldsymbol{q} \in \mathcal{C}, \boldsymbol{D}(\boldsymbol{q}) = \boldsymbol{D}\}$$

and the inner value function

$$W^{\mathrm{RA}(\boldsymbol{D})} = \max_{\boldsymbol{q}:\boldsymbol{q}\in\mathcal{C}, \boldsymbol{D}(\boldsymbol{q})=\boldsymbol{D}} \left\{ \sum_k \psi_{B_k}\big(U_{k(\boldsymbol{q})}\big) + \sum_{j\notin \text{ MM}} w_j q_j \right\}$$

for $\boldsymbol{D} \in \mathcal{D}$, extended by $-\infty$ outside $\mathcal{D}$. Then W^{RA} is concave.

Proof. Let $\boldsymbol{D}^1, \boldsymbol{D}^2 \in \mathcal{D}$ and let $\boldsymbol{q}^1$ and $\boldsymbol{q}^2$ attain the maxima defining $W^{\mathrm{RA}(\boldsymbol{D}^1)}$ and $W^{\mathrm{RA}(\boldsymbol{D}^2)}$. For any $\theta \in [0,1]$, convexity of $\mathcal{C}$ gives $\boldsymbol{q}^\theta = \theta \boldsymbol{q}^1 + (1-\theta)\boldsymbol{q}^2 \in \mathcal{C}$. Linearity of $\boldsymbol{D}$ and $\boldsymbol{U}$ gives

$$\boldsymbol{D}\big(\boldsymbol{q}^\theta\big) = \theta \boldsymbol{D}^1 + (1-\theta)\boldsymbol{D}^2, \quad \boldsymbol{U}\big(\boldsymbol{q}^\theta\big) = \theta \boldsymbol{U}\big(\boldsymbol{q}^1\big) + (1-\theta)\boldsymbol{U}\big(\boldsymbol{q}^2\big)$$

so $\boldsymbol{q}^\theta$ is feasible for $\theta\boldsymbol{D}^1 + (1-\theta)\boldsymbol{D}^2$. The inner objective

$$G(\boldsymbol{q}) = \sum_k \psi_{B_k}\big(U_{k(\boldsymbol{q})}\big) + \sum_{j\notin \text{ MM}} w_j q_j$$

is concave in $\boldsymbol{q}$, so

$$W^{\mathrm{RA}(\theta\boldsymbol{D}^1+(1-\theta)\boldsymbol{D}^2)} \geq G\big(\boldsymbol{q}^\theta\big) \geq \theta G\big(\boldsymbol{q}^1\big) + (1-\theta) G\big(\boldsymbol{q}^2\big)$$

which equals $\theta W^{\mathrm{RA}(\boldsymbol{D}^1)} + (1-\theta) W^{\mathrm{RA}(\boldsymbol{D}^2)}$. □

5.3. Proof

(1) Concavity, existence, and lift. By Proposition 3, each ψ_{B_k} is concave. Since U_k and $\boldsymbol{D}$ are linear in $\boldsymbol{q}$, the reduced-form objective of Theorem 5 is concave on the compact polytope $\mathcal{C}$, so an optimum exists by Weierstrass. The deployed-value lift follows from Proposition 4, and at any optimum the lifted variable is $V_k^* = \max(U_k, B_k)$ with

$$\alpha_k = B_k / V_k^* = \psi_{(B_k)'}(U_k) \leq 1$$

(2) Limit order exactness and two regimes. Work in the equivalent deployed-value lift of Proposition 4. Let $\eta_k \geq 0$ be the multiplier for the constraint $U_{k(\boldsymbol{q})} - V_k \leq 0$. Stationarity with respect to V_k gives

$$B_k / V_k - 1 + \eta_k = 0$$

so $\eta_k = 1 - \alpha_k$. The KKT condition for fill q_i of MM k is then

$$\alpha_k L_i - p_{m(i)} = \lambda_i^+ - \lambda_i^-$$

because the objective contributes $+L_i$ through U_k, while the active-value constraint subtracts $\eta_k L_i$. A fill ($q_i > 0$) requires $\alpha_k L_i \geq p_{m(i)}$, hence $L_i \geq p_{m(i)}/\alpha_k \geq p_{m(i)}$: the limit price must exceed the clearing price. No negative-welfare fill is possible.

The two regimes are immediate from $V_k^* = \max(U_k, B_k)$. If $U_k > B_k$, then $V_k = U_k$, so $s_k = 0$ and $\alpha_k = B_k/U_k < 1$. If $U_k < B_k$, then $V_k = B_k$, so $s_k = B_k - U_k > 0$ and $\alpha_k = 1$. In the over-capitalized regime, the fill condition $L_i \geq p_{m(i)}$ is exactly the risk-neutral UCP condition — the MM clears identically to the LP.

(3) Budget absorption. Multiply the fill KKT by q_i and sum over $i \in \mathrm{MM}_k^+$. By complementary slackness ($\lambda_i^- q_i = 0$):

$$\underbrace{\alpha_k \sum_{i \in \mathrm{MM}_k^+} L_i q_i}_{=U_k} = \sum_{i \in \mathrm{MM}_k^+} p_{m(i)} q_i + \underbrace{\sum_{i \in \mathrm{MM}_k^+} \lambda_i^+ q_i}_{\geq 0}$$

So $\sum p_{m(i)} q_i \leq \alpha_k U_k$. Since $\alpha_k V_k = B_k$ and $s_k = V_k - U_k$, we have

$$\alpha_k U_k = B_k - \alpha_k (V_k - U_k) = B_k - \alpha_k s_k \leq B_k$$

and therefore

$$\sum_{i \in \mathrm{MM}_k^+} p_{m(i)}^* q_i^* + s_k^* \leq B_k$$

Each MM's total deployment — capital on fills plus retained cash — is at most B_k. (The gap $\sum \lambda_i^+ q_i$ is infra-marginal surplus from fully-filled orders — profit that does not require additional capital.) The budget emerges from the objective, not from an explicit constraint. This is the Eisenberg-Gale mechanism extended to quasi-linear utilities: the ln singularity absorbs the budget, and the retained-cash variable absorbs the surplus.

(4) Price extraction and uniqueness for $b > 0$. For $b > 0$, prices are the gradient of the smoothed minting cost:

$$p_k = \partial C_b / \partial D_k = \mathrm{softmax}(\boldsymbol{D}/b)$$

By Proposition 5, the demand-space value function W^{RA} is concave. Therefore the primal problem is

$$\max_{\boldsymbol{D}} \left[W^{\mathrm{RA}(\boldsymbol{D})} - C_{b(\boldsymbol{D})} \right]$$

and Fenchel-Rockafellar duality gives the price problem

$$\min_{\boldsymbol{p} \in \Delta} [W_{\mathrm{RA}}^*(\boldsymbol{p}) + C_b^*(\boldsymbol{p})]$$

where W_{RA}^* is convex and $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k$ is strictly convex on Δ by Theorem 2. Hence the dual objective is strictly convex on the simplex, so the minimizing price vector is unique. At $b = 0$, prices are the LP dual variables of the epigraph constraints $M \geq D_k$, which may be non-unique when multiple outcomes tie at $\max_j D_j$.

(5) Polynomial solvability. At $b = 0$, Proposition 9 gives an exponential-cone formulation of P_0^{RA}, so standard interior-point solvers apply in polynomial time. For $b > 0$, the smoothed minting cost likewise admits a conic lift with $O(K)$ additional exponential cones (see Conic Formulation below), or can be handled directly by smooth convex optimization. □

5.4. Price Dual and Temperature

Proposition 6 (Risk-Averse Price Dual). *Let W^{RA} be the concave demand-space value function of Proposition 5. Then the clearing-price problem is*

$$\min_{\boldsymbol{p} \in \Delta} [W_{\mathrm{RA}}^*(\boldsymbol{p}) + C_b^*(\boldsymbol{p})]$$

where $C_b^*(\boldsymbol{p}) = b \sum_k p_k \ln p_k$ for $b > 0$ and $C_0^* = \delta_\Delta$. Equivalently, at $b = 0$ the price set is the minimizer set of W_{RA}^* over the simplex. For $b > 0$, the minimizer is unique.

Proof. This is exactly the dual characterization from part (4) of Theorem 5 together with Theorem 1 and Theorem 2. For $b > 0$, uniqueness follows from strict convexity of the entropy term on Δ. □

Example. Binary market ($K = 2$), $b \to 0$. An MM ($B = 50$) posts two buy-Yes orders: A at $L_A = 0.70$, qty 100; B at $L_B = 0.55$, qty 100. A retail trader posts buy-No at $L_C = 0.60$, qty 200.

LP (no budgets). All fill at balanced minting ($D_{\text{Yes}} = D_{\text{No}} = 200$). Welfare = 45. MM capital at any supporting price exceeds B: even at the lowest dual price $p_{\text{Yes}} = 0.40$, capital $= 0.40 \times 200 = 80 > B$.

Risk-averse. The MM is capital-constrained ($U^{\text{LP}} = 125 > B = 50$, $s = 0$). Only order A fills: $q_A = 100$, $q_B = 0$, $q_C = 100$. The retail order is partially filled, pinning $p_{\text{No}} = L_C = 0.60$, hence $p_{\text{Yes}} = 0.40$. The capital-scarcity factor is $\alpha = B/U = 50/70 \approx 0.71$. Order A fills: $\alpha L_A = 0.50 > p_{\text{Yes}} = 0.40$. Order B does not fill: $\alpha L_B = 0.39 < 0.40$ — the capital-scarcity factor throttles the lower-ROI order. Welfare = 30. Capital consumed $= 0.40 \times 100 = 40 \le B$, with the gap of 10 being infra-marginal surplus from order A.

Welfare gap. Actual gap = 15; exact bound $= \Delta - B\ln(1 + \Delta/B) \approx 29$ (where $\Delta = U^{\text{LP}} - B = 75$). The bound is conservative because the retail side contracts proportionally.

In the risk-neutral model, entropy smoothing was the only source of concavity — insufficient against budget non-convexity (Proposition 2). Here, the reduced-form MM utility provides curvature at every temperature, and the deployed-value lift of Proposition 4 makes that curvature explicit. Even at $b = 0$ (pure LP minting cost), the program remains a well-posed concave optimization problem:

$$P_0^{\text{RA}} : \max_{\boldsymbol{q} \in \mathcal{C}, \boldsymbol{V} \ge \boldsymbol{U}(\boldsymbol{q})} \sum_k \left[B_k \ln V_k - V_k + U_{k(\boldsymbol{q})} \right] + \sum_{j \notin \text{ MM}} w_j q_j - \max_k D_k$$

The temperature dependence is therefore:

Property	$\boldsymbol{b > 0}$	$\boldsymbol{b = 0}$
Concave formulation and existence	Yes	Yes
Reduced-form utility / budget absorption	Yes	Yes
Exact limit-order KKT conditions	Yes	Yes
Clearing prices	Unique softmax	Simplex-valued LP duals; may be non-unique
Conic solvability	Yes	Yes

The $-\max_k D_k$ term is concave (not strictly), and the log term continues to regularize capital deployment. So existence, limit-order exactness, budget absorption, and conic solvability all survive at $b = 0$. What fails at zero temperature is uniqueness: the entropy term disappears from Proposition 6, prices become LP-type dual variables, and multiple outcomes can tie at $D_k = \max_j D_j$. Any $b > 0$ restores a strictly convex price dual, with subsidy bounded by $b \ln K$ (Proposition 1).

5.5. Consequences of the Affine Envelope

The reduced-form utility of Proposition 3 is exactly an affine LP objective minus a nonnegative penalty on the capital-binding region. Define the affine-envelope gap

$$g_{B(U)} = U + B\ln B - B - \psi_{B(U)} = \begin{cases} 0 & \text{if } U \le B \\ U - B - B\ln(U/B) & \text{if } U \ge B \end{cases}$$

Then $g_{B(U)} \geq 0$ for all U, with equality exactly on the slack-budget region. For any feasible fill vector,

$$\sum_k \psi_{B_k}\big(U_{k(\boldsymbol{q})}\big) = \sum_k \big[U_{k(\boldsymbol{q})} + B_k \ln B_k - B_k\big] - \sum_k g_{B_k}\big(U_{k(\boldsymbol{q})}\big)$$

So, up to the constant $\sum_k (B_k \ln B_k - B_k)$, the reduced-form clearing objective is the LP objective minus the envelope gap.

Proposition 7 (LP Recovery). *Let $\boldsymbol{q}^{\mathrm{LP}}$ be any unconstrained LP optimum (P_b without budgets). If $B_k \geq U_k^{\mathrm{LP}} = \sum_{i \in \mathrm{MM}_k^+} L_i q_i^{\mathrm{LP}}$ for every MM k, then $\boldsymbol{q}^{\mathrm{LP}}$ is optimal for the reduced-form risk-averse program. In the deployed-value lift, the corresponding retained cash is $s_k = B_k - U_k^{\mathrm{LP}}$. In particular, the LP and risk-averse programs share an optimum whenever all MM budgets are non-binding.*

Proof. If $B_k \geq U_k^{\mathrm{LP}}$ for all k, then $g_{B_k}\big(U_k^{\mathrm{LP}}\big) = 0$ for every MM. So at $\boldsymbol{q}^{\mathrm{LP}}$ the reduced-form objective equals the LP objective plus the constant $\sum_k (B_k \ln B_k - B_k)$. For any other feasible $\boldsymbol{q}$, the reduced-form objective is that same LP objective minus the non-negative penalty $\sum_k g_{B_k}\big(U_{k(\boldsymbol{q})}\big)$. Since $\boldsymbol{q}^{\mathrm{LP}}$ maximizes the LP objective, it also maximizes the reduced-form objective. The corresponding deployed values are $V_k = B_k$, equivalently $s_k = B_k - U_k^{\mathrm{LP}}$. □

The retained-cash variable s_k acts as a *numeraire good*, eliminating the over-fill pathology of the naive $B_k \ln U_k$ model (which forces all budget into fills, including unprofitable ones). The quasi-linear structure is essential: without s_k, over-capitalized MMs would distort fills to exhaust their budget.

Proposition 8 (Welfare Gap Bound). *Let W^{LP} and W^{RA} denote the total welfare (LP objective value) at the LP and risk-averse optima respectively, and let $\Delta_k = \max\big(0, U_k^{\mathrm{LP}} - B_k\big)$ be MM k's budget shortfall. Then:*

$$W^{\mathrm{LP}} - W^{\mathrm{RA}} \leq \sum_k \left[\Delta_k - B_k \ln\left(1 + \frac{\Delta_k}{B_k}\right)\right] \leq \sum_k \frac{\Delta_k^2}{2B_k}$$

Over-capitalized MMs ($\Delta_k = 0$) contribute nothing. The bound is quadratic in the shortfall.

Proof. Write $F(\boldsymbol{q})$ for the LP welfare objective (the unconstrained batch-clearing objective with the same minting cost C_b). By the envelope decomposition above, the reduced-form objective is

$$F(\boldsymbol{q}) + \sum_k (B_k \ln B_k - B_k) - \sum_k g_{B_k}\big(U_{k(\boldsymbol{q})}\big)$$

Since $\boldsymbol{q}^{\mathrm{RA}}$ maximizes this penalized objective,

$$F\big(\boldsymbol{q}^{\mathrm{RA}}\big) - \sum_k g_{B_k}\big(U_{k(\boldsymbol{q}^{\mathrm{RA}})}\big) \geq F\big(\boldsymbol{q}^{\mathrm{LP}}\big) - \sum_k g_{B_k}\big(U_k^{\mathrm{LP}}\big)$$

Rearranging and using $g_B \geq 0$ gives

$$W^{\mathrm{LP}} - W^{\mathrm{RA}} \leq \sum_k g_{B_k}\big(U_k^{\mathrm{LP}}\big)$$

If $U_k^{\mathrm{LP}} = B_k + \Delta_k$, then

$$g_{B_k}\big(U_k^{\mathrm{LP}}\big) = \Delta_k - B_k \ln\left(1 + \frac{\Delta_k}{B_k}\right)$$

while over-capitalized MMs have $\Delta_k = 0$ and contribute zero. The quadratic bound follows from $\ln(1+x) \geq x - \frac{x^2}{2}$. □

The bound is tight when all orders are marginal ($L_i \approx p_k$) and conservative for real order ladders, where spread-out limit prices give $V_k^{\mathrm{RA}} \gg B_k$. The exact expression grows sublinearly in Δ_k: the throttled fills are precisely the least profitable ones.

5.6. Fisher Interpretation

In a Fisher market (Eisenberg & Gale 1959), n consumers with budgets B_k purchase divisible goods with fixed supply s_j at prices p_j. A market equilibrium can be characterized as a solution to:

$$\mathrm{EG}: \quad \max_{\boldsymbol{x} \geq 0} \quad \sum_k B_k \ln U_k(\boldsymbol{x}_k) \quad \text{s.t.} \quad \sum_k x_{kj} \leq s_j \quad \forall j$$

where $U_{k(\boldsymbol{x}_k)} = \sum_j u_{kj} x_{kj}$ is consumer k's linear utility over goods. The supply constraints have dual variables p_j — the equilibrium prices. Budget constraints do not appear explicitly; they emerge from the $B_k \ln U_k$ objective by the same telescoping argument as our proof of (3). Adding a cash variable s_k with cost $-s_k$ yields the *quasi-linear* Fisher market (Cole et al. 2017, Chen et al. 2009) — agents can retain unspent budget as cash rather than being forced to spend it all on goods. Quasi-linear Fisher markets are well-studied: equilibrium existence and polynomial-time computation are known under mild conditions, while uniqueness typically concerns prices or utility profiles under additional regularity.

Program P_b^{RA} is a quasi-linear Fisher market with two extensions:

Component	**Fisher market (EG)**	**Batch auction (P_b^{RA})**
Consumers	n agents with budgets B_k	MMs with budgets B_k
Goods	Divisible commodities	Outcome shares
Supply	Fixed endowment s_j	Endogenous: minting at cost C_b
Utility	$U_k = \sum_j u_{kj} x_{kj}$	$U_k = \sum_{i \in \mathrm{MM}_k^+} L_i q_i$
Cash	Optional ($s_k \geq 0$)	Retained budget ($s_k \geq 0$)
Prices	Dual of $\sum_k x_{kj} \leq s_j$	Gradient of C_b (softmax)

The prediction market extends the quasi-linear Fisher market in two ways: (1) supply is endogenous — shares are created by minting at cost $C_{b(\boldsymbol{D})}$ rather than drawn from a fixed endowment, and (2) non-MM ("retail") orders contribute linear welfare alongside the log-utility MMs. Both extensions preserve concavity.

5.7. What Changes and What Doesn't

What changes. Under P_b^{RA}, capital-constrained MMs ($\alpha_k < 1$) prioritize their highest-ROI fills rather than filling every profitable order equally. An MM concentrating capital on one market faces the ln penalty: the first dollar of fill generates high marginal utility, additional dollars generate diminishing utility. This naturally diversifies MM capital across markets. Over-capitalized MMs ($\alpha_k = 1$) behave identically to the LP — the log model only affects sizing when the budget actually binds.

What doesn't change. Non-MM orders are still matched by UCP at clearing prices. The minting mechanism is unchanged. Prices are still softmax (for $b > 0$) or LP duals (for $b = 0$). Limit orders are exact: no order fills below its stated price.

5.8. Buy/Sell Reduction

In each binary market, MM positions are netted per outcome before clearing: opposing buys and sells cancel, leaving a net directional position. Since "sell Yes at L" is economically equivalent to "buy No at $1 - L$" (with capital cost $1 - L$ per share), all net positions are expressed as buys. These net buy orders form MM_k^+ and their capital costs enter the budget accounting through U_k. The only MM sell orders that bypass the budget are liquidations of existing inventory — shares already held, not newly created short positions. Per outcome per batch, the MM is either net buying or net selling, never both.

5.9. Multiple Groups and Bundle Orders

Nothing in Theorem 5 requires a single mutually exclusive group. Consider N groups, each with K_j mutually exclusive outcomes. The joint state space is $\mathcal{S} = \prod_j \{1, ..., K_j\}$ (exactly one joint state is realized). Bundle orders — orders whose payoffs depend on the joint state — create demand D_s over $\mathcal{S}$. The minting cost generalizes to $V(\boldsymbol{D}) = \max_s D_s$ (minting one complete set of all joint states costs \$1), and the smoothed version is $C_b = b \ln \sum_s \exp(D_s/b)$. Both are convex in $\boldsymbol{D}$, and $\boldsymbol{D}$ is linear in $\boldsymbol{q}$. The proof of Theorem 5 goes through unchanged for any fixed state space representation: each MM term $B_k \ln V_k - V_k + U_k$ remains concave, $-C_b$ remains concave, and budget absorption and limit order exactness still hold. The open question is whether the program can be solved without explicitly enumerating the joint states (see Open Problems).

When no orders span multiple groups, the joint problem decomposes: $C_b = \sum_j C_b^{G_j}$ (the product-LMSR factorization of Chen and Pennock 2007). Cross-group orders break this separability, coupling groups transitively — if orders link groups A–B and B–C, the clearing problem requires the full $K_A \times K_B \times K_C$ state space. The mathematical obstruction is purely computational, not structural; the companion decomposition note proves exact componentwise decomposition when the coupling graph splits and analyzes mirror-descent coordination of shared MM budgets.

6. Computation

6.1. Conic Formulation

At $b = 0$, the minting cost $\max_k D_k$ is modeled by LP epigraph constraints ($M \geq D_k$). In the deployed-value lift of Proposition 4, all nonlinearity is then isolated in the terms $\ln V_k$: after introducing epigraph variables $t_k \leq \ln(V_k)$, the remaining objective and constraints are linear. Each constraint $t_k \leq \ln(V_k)$ is modeled by a single exponential cone: $(t_k, 1, V_k) \in \mathcal{K}_{\exp} = \{(x, y, z) : y \exp(x/y) \leq z\}$. The full program is a conic optimization problem solvable by standard interior-point solvers (Clarabel, MOSEK).

Proposition 9 (Conic Reformulation). *At $b = 0$, P_0^{RA} is equivalent to a conic program: minimize a linear objective over LP constraints and n_{MM} exponential cone constraints (one per MM). Clearing prices are the dual variables of the minting epigraph constraints. For $b > 0$, the log-sum-exp minting cost adds $O(K)$ additional exponential cones (one per outcome); the augmented-LP approach below avoids this overhead.*

Remark (Augmented LP). The EG Hessian $H = -\sum_k (B_k/V_k^2) \ell_k \ell_k^T$ has rank at most n_{MM} (the number of MMs), where ℓ_k is MM k's welfare weight vector. By Sherman–Morrison–Woodbury, each Newton step costs the same as an LP interior-point step plus an $O(n \cdot n_{\mathrm{MM}})$ rank-n_{MM} correction — dominated by the $O(n)$ LP cost when $n_{\mathrm{MM}} \ll n$. The EG problem has the same asymptotic complexity as the LP.

6.2. Decomposition for Exponential State Spaces

When no orders span multiple market groups, the minting cost separates: $C_b = \sum_j C_b^{G_j}$ (product-LMSR factorization). This suggests a decomposition of the EG program into independent per-group subproblems, coordinated by MM budget allocation.

With log utility, splitting an MM's budget B_k across components m leads to a smooth concave coordination problem: the coordination objective $\max \sum_m W_m^*(\boldsymbol{B}^m)$ subject to $\sum_m B_k^m = B_k$ has gradient $\partial W_m^* / \partial B_k^m = \ln V_k^{m*}$ (envelope theorem), where $V_k^m = U_k^m + s_k^m$. The first-order condition equalizes these component values across active components. Standard mirror descent (multiplicative weights on budget shares) is then a natural algorithmic candidate, with each iteration requiring only independent per-component solves. A companion note develops this decomposition in detail.

With linear welfare, the same budget allocation is combinatorial: a hard constraint $\text{cap}_k \leq B_k$ cannot be smoothly split across components. This is the computational payoff of the Fisher market structure — the same log-utility mechanism that absorbs budget constraints in the monolithic program also enables decomposition across components.

Validation. An implementation of the LP, conic, and EG solvers confirms the theoretical predictions on synthetic order books with 50 market groups, $\sim 11,000$ single-market orders, and 3 budget-constrained MMs. The quasi-Fisher (conic) and linear (LP) solvers produce near-identical welfare (gap $< 0.7\%$ across all budget scales), consistent with Proposition 7 and Proposition 8. When budgets are non-binding, the programs share an optimum. Decomposition with mirror descent on budget shares converges toward the monolithic solution on single-market order books, as expected from the product-LMSR factorization.

Current timing benchmarks should be read in regime context. On independent binary markets, monolithic LP clearing is already cheap because the state space is small, so decomposed solving pays the overhead of repeated coordination iterations without gaining asymptotic advantage. In those tests, decomposed LP nearly matches monolithic welfare but is somewhat slower wall-clock, while decomposed conic solving is currently limited by numerical conditioning when some component budgets become very small. The decomposition architecture is aimed at the opposite regime: bundle orders and coupled market groups, where monolithic enumeration becomes exponential and coordination overhead is the cheaper problem.

Concrete regime picture: if the bundle-coupling graph on 20 binary groups splits into four connected components of five groups each, monolithic joint-state clearing sees $2^{20} \approx 10^6$ states, while componentwise clearing sees only $4 \cdot 2^5 = 128$ states plus a smooth budget-coordination layer. That is the setting decomposition is for. The current benchmarks do not yet probe that regime.

7. Discussion

7.1. Main Contributions

1. **Reduced-form MM utility.** The paper identifies the exact MM utility induced by budget plus retained cash: affine when capital is slack and logarithmic when capital binds (Proposition 3). The deployed-value lift Proposition 4 is the exact derivation and solver form behind this reduced form.

2. **Observable market outcomes.** The risk-averse program has exact limit-order KKT conditions, endogenous budget absorption, and an explicit price dual Proposition 6. For every $b > 0$, the dual is strictly convex, so clearing prices are unique; at $b = 0$, the remaining ambiguity is the familiar LP dual degeneracy.

3. **Algorithmic consequence.** The formulation admits exponential-cone representations and an augmented-LP interpretation (Proposition 9). Decomposition is most useful in the coupled multi-group regime where the joint state space is exponential, not in the small independent-market benchmarks where monolithic LP is already cheap.

7.2. Open Problems

1. **Welfare gap tightness.** The quadratic bound of Proposition 8 is conservative for typical order books: synthetic experiments show gaps under 0.7% even at $10 \times$ budget oversubscription. Tighter instance-dependent bounds — exploiting the structure of real MM ladder orders — would sharpen the practical guarantees.

2. **Efficient combinatorial clearing.** The Fisher market isomorphism extends to joint state spaces (see Multiple Groups and Bundle Orders), but the state space $|\mathcal{S}| = \prod K_j$ is exponentially large. Cross-group bundle orders couple groups transitively: if orders span A–B and B–C, clearing requires the full $K_A \times K_B \times K_C$ space. In practice, a handful of bundle orders can connect all groups. Each order's payoff is a low-rank tensor (touching ≤ 5 groups), so the demand D_s has exploitable structure. Can the EG program be solved in time polynomial in the number of orders rather than the state space?

3. **Risk-averse observables at $b = 0$.** The reduced-form program and its deployed-value lift prove existence, limit-order exactness, and budget absorption at every temperature, and unique clearing prices for every $b > 0$. What remains open is the zero-temperature edge case: under what additional regularity are prices, aggregate demands, or projected fills unique when the minting cost is $\max_k D_k$ and the LP dual is set-valued?

7.3. Connection to Prior Work

Eisenberg and Gale (1959): The convex program for Fisher market equilibrium. Our Theorem 5 extends their framework to prediction markets with endogenous supply (minting) and mixed agent types (log-utility MMs alongside linear-welfare retail), with reduced-form utility Proposition 3 and deployed value V_k as the bridge variables.

Abernethy, Chen, and Vaughan (2013): Axiomatic foundation for cost-function market makers. Our §2 recasts their framework as batch-auction clearing: $V = \max_k D_k$ is the simplest cost function, LMSR its entropy smoothing.

Breiman (1961): Optimality of the Kelly criterion for repeated games. §4 uses this as a repeated-game motivation for diminishing returns, not as a literal behavioral theorem about every MM.

Devanur and Dudík (2015): Budget constraints and additivity for sequential LMSR. Their budget additivity hints at hidden convexity in the risk-neutral setting. Our contribution is a concave EG-type formulation for batch auctions; whether risk-neutral batches also have hidden convexity remains open.

Fortnow, Kilian, Pennock, and Wellman (2005): LP for combinatorial call markets. Our group minting encodes mutual exclusivity in $O(K)$ constraints vs $O(2^K)$ enumeration.

Chen and Pennock (2007): Bounded-loss market makers and the product-LMSR factorization. The parameter b is their loss bound; $b = 0$ is achievable in batch clearing by complementary slackness.

Cole et al. (2017), Chen et al. (2009): Quasi-linear Fisher markets. Our P_b^{RA} is structurally a quasi-linear Fisher market; the contribution is the deployed-value /

reduced-form-utility view for prediction markets with endogenous supply and LMSR pricing.

Budish, Cramton, and Shim (2015): Frequent Batch Auctions for equity markets. Our framework applies FBAs to prediction markets.

Implementation and experimental code available at the project repository. All solvers (LP, EG, conic) and the decomposition framework are implemented in Rust.